

Bank Loan Portfolio Overview and Credit Risk & NPL Tracker

Power BI Project Report | Whole-Bank Analysis | Synthetic Retail Banking Data

Prepared for portfolio documentation and LinkedIn/GitHub project presentation

1,000 Loan Accounts portfolio size	\$50.71M Total Disbursed origination volume	\$22.06M Outstanding Balance current exposure	12.79% Avg. Interest Rate portfolio yield proxy	\$50.71K Avg. Loan Size per account
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21.18% NPL Ratio 90+ DPD balance / outstanding	17.00% Default Rate defaulted accounts / total accounts	\$4.673M NPL Balance 90+ DPD exposure	\$4.67M High-Risk Exposure risk-rated high exposure
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1. Executive Summary

This project converts a synthetic retail banking dataset into two Power BI reporting layers: a **Bank Loan Portfolio Overview** for business performance and a **Credit Risk & NPL Tracker** for risk monitoring. The model contains 1,000 loan accounts with total original disbursement of \$50.71M and current outstanding exposure of \$22.06M. The portfolio is balanced across major retail lending products, with Auto Loan as the largest product by disbursement volume at \$9.44M, followed by Home Loan at \$9.10M and Personal Loan at \$8.86M.

- **Overall portfolio scale:** The bank has \$22.06M outstanding against \$50.71M originally disbursed, indicating that approximately 43.5% of original disbursement remains as active exposure.
- **Branch performance:** Orlando Central is the largest production branch, with \$6.8M disbursed, 125 loans, and \$3.32M outstanding. Tampa Bay has the smallest loan count at 80 and the lowest disbursement volume at roughly \$4.1M.
- **Credit quality:** The whole-bank NPL ratio is 21.18%, driven by \$4.673M of 90+ DPD exposure. The account-based default rate is 17.00%, meaning roughly 170 out of 1,000 accounts are classified as defaulted under the project rule.
- **Risk concentration:** Tampa Bay has the highest NPL ratio at 31.2%, followed by Houston South at 30.2%, Orlando Central at 26.0%, and Dallas Central at 25.4%. Education Loan has the highest product default rate at 20.26%.
- **Provisioning concern:** The simplified provision amount of \$7.597K is extremely small relative to \$4.673M of NPL balance, producing an implied coverage ratio near 0.16%. This should be treated as a model/data-calibration issue and corrected before using the report for provision adequacy conclusions.

2. Project Scope, Data Model, and Methodology

Scope. The project is built on synthetic/mock data designed to resemble common retail bank loan portfolio fields. It is intended for learning, portfolio demonstration, and Power BI practice only. It does not contain real customer or bank data.

Data model. The Power BI model uses a star-schema structure: Fact_Loans is the central transaction/risk fact table, while Dim_Branch, Dim_Product, Dim_Customer, and Date_Table provide filtering and grouping. This design lets users analyze risk and performance by region, branch, loan type, credit score band, loan status, delinquency bucket, and year.

Risk logic. For this project, a loan is treated as non-performing when Days Past Due is 90 or greater. This is a simplified learning assumption that aligns with common supervisory reporting logic for severely past-due loans. The provision calculation is also simplified and should be treated as a mock rule, not a production expected-credit-loss model.

3. Key KPI Definitions Used in the Dashboard

KPI / Term	Meaning and Analytical Use
Outstanding Balance	Current amount is still owed by borrowers; this is the key exposure base for risk analysis.
Days Past Due (DPD)	Number of days a scheduled payment is late; used to identify delinquency severity.
Delinquency Bucket	DPD grouped into Current, 1-29, 30-59, 60-89, and 90+ DPD for easier risk monitoring.
NPL Balance	Outstanding balance of loans in the 90+ DPD bucket.
NPL Ratio	NPL Balance divided by Total Outstanding Balance; a core asset-quality measure.
Default Rate	Defaulted loan accounts divided by total loan accounts; an account-based risk frequency measure.
High-Risk Exposure	Outstanding balance of loans tagged as High risk; used to prioritize monitoring and collections.
Provision Coverage Ratio	Provision amount divided by NPL balance; indicates reserve coverage against severe delinquency exposure.
Cost-to-Income Ratio	Operating expense divided by operating income; an efficiency KPI rather than a pure credit-risk KPI.

4. Bank Loan Portfolio Overview - Whole-Bank Analysis

Portfolio volume and exposure. The dashboard shows 1,000 loans, \$50.71M in total disbursement, \$22.06M in current outstanding balance, and an average loan size of \$50.71K. The average interest rate is 12.79%, with branch-level averages concentrated in a narrow band from 12.27% to 13.40%. This suggests that the simulated portfolio has relatively consistent pricing across branches, while exposure differences are mainly driven by volume and loan size rather than sharply different interest rates.

Branch production. Orlando Central is the strongest originator, with \$6.8M disbursed and 125 accounts. Atlanta Midtown and Houston South follow at approximately \$5.5M and \$5.4M. Tampa Bay is the smallest production branch at approximately \$4.1M disbursed and 80 accounts. From a business-performance view, Orlando Central is the growth leader; from a risk view, its large exposure means any deterioration in credit quality has a bigger portfolio impact.

Product mix. The product portfolio is fairly diversified. Auto Loan is the largest product by original disbursement at \$9.44M or 18.62% of the portfolio. Home Loan contributes \$9.10M, Personal Loan contributes \$8.86M, Small Business Loan contributes \$8.54M, Education Loan contributes \$7.72M, and Credit Card Loan contributes \$7.04M. Because no single product dominates the full book, risk monitoring should compare both product default rates and product exposure size.

Portfolio Area	Highest / Key Value	Analytical Reading	Action
Branch volume	Orlando Central: \$6.8M disbursed	Largest production branch and largest account count.	Monitor risk closely because high volume amplifies losses.
Outstanding exposure	Orlando Central: \$3.319M outstanding	Highest current branch exposure.	Compare with branch NPL ratio and delinquency mix.
Average interest rate	Dallas Central: 13.40%	Highest branch average rate.	Check if higher yield compensates for higher risk.
Product volume	Auto Loan: \$9.44M	Largest product share at 18.62%.	Track product-level delinquency and default trends.

5. Credit Risk & NPL Tracker - Deep Dive

Overall credit quality. The whole-bank NPL ratio is 21.18%, based on \$4.673M of 90+ DPD exposure divided by \$22.059M of total outstanding balance. The default rate is 17.00%, indicating that approximately 17% of loan accounts are flagged as defaulted. Because this is synthetic data, the absolute values should not be benchmarked against a real institution. However, within this project, the values indicate a meaningfully stressed portfolio that requires branch/product-level investigation.

6. Delinquency Composition

Bucket	Balance	Share of Outstanding	Interpretation
Current	\$8,212,827	37.23%	Healthy book but still needs monitoring for early warning movement.
1-29 DPD	\$2,258,738	10.24%	Early delinquency; reminder and borrower contact stage.
30-59 DPD	\$4,703,518	21.32%	Material watchlist exposure; collections should intervene.
60-89 DPD	\$2,210,935	10.02%	Serious delinquency and likely pre-NPL migration risk.
90+ DPD	\$4,672,576	21.18%	Non-performing exposure under project definition.

Key insight. Only 37.23% of outstanding exposure is current. The 30-59 DPD bucket is \$4.704M, slightly higher than the 90+ DPD NPL bucket of \$4.673M. This is important because 30-59 DPD loans represent a forward-looking risk pipeline: if collections are not effective, part of this bucket may roll into 60-89 and 90+ DPD.

7. Branch and Product Risk Concentration

Branch risk. Tampa Bay has the highest NPL ratio at 31.2%, followed by Houston South at 30.2%, Orlando Central at 26.0%, Dallas Central at 25.4%, and Nashville West at 24.5%. Miami Downtown has the lowest NPL ratio at 9.0%. The risk story is not only about loan volume: Tampa Bay is the smallest production branch by disbursement, yet it has the highest NPL ratio. This suggests that risk quality, borrower mix, or collections performance can be more important than volume alone.

Product risk. Education Loan has the highest default rate at 20.26%, followed by Credit Card Loan at 17.57%, Personal Loan at 17.44%, Small Business Loan at 16.07%, Home Loan at 15.91%, and Auto Loan at 15.30%. The difference between the highest and lowest product default rate is roughly 4.96 percentage points, meaning risk is broad-based rather than concentrated in only one product. Education Loans should still be reviewed first because they combine the highest default frequency with an unsecured or weaker-collateral risk profile in many retail-lending contexts.

Risk Lens	Highest-Risk Segment	Metric	Management Question
Branch	Tampa Bay	31.2% NPL Ratio	Is this driven by borrower quality, product mix, or collections delay?
Branch	Houston South	30.2% NPL Ratio	Does the branch need underwriting/collections review?
Product	Education Loan	20.26% Default Rate	Should credit policy, pricing, or borrower screening be adjusted?
Bucket	30-59 DPD	\$4.704M	How much will roll forward into 60+ and 90+ DPD?

8. Time View, Provisioning, and Model Limitations

Time analysis caution. The dashboard currently uses the loan disbursement/origination date as the date relationship. Therefore, the monthly credit-risk charts are best interpreted as vintage-style views: they show risk by the month loans were originated, not necessarily the true month-by-month movement of borrower performance. For a production-grade credit-risk dashboard, add a monthly loan performance snapshot table with one row per loan per reporting month.

NPL and provision view. The highest NPL origination months include 2026-Apr at \$416.5K, 2025-Nov at \$316.2K, 2026-May at \$290.0K, and 2025-May at \$263.7K. These months should be investigated for underwriting changes, borrower mix, seasonal effects, and product concentration. However, the provision amount shown in the dashboard is far too low relative to NPL exposure. The provision calculation should be recalibrated using outstanding balance and risk severity or replaced with a simplified expected loss approach: $EAD \times PD \times LGD$, where EAD is exposure at default, PD is probability of default, and LGD is loss given default.

9. Conclusion

This Power BI project demonstrates the transition from descriptive banking analytics to credit-risk monitoring. The Bank Loan Portfolio Overview explains how lending is distributed across branches and products. The Credit Risk & NPL Tracker then evaluates the quality of that exposure through delinquency buckets, NPL ratio, default rate, high-risk exposure, and branch/product risk concentration. The most important analytical finding is that the portfolio has a meaningful severe-delinquency burden, with 21.18% of outstanding balance in 90+ DPD and 17.00% of accounts flagged as defaulted. The second major finding is that provision coverage appears materially under-calibrated and should be corrected before any reserve adequacy conclusion is made.